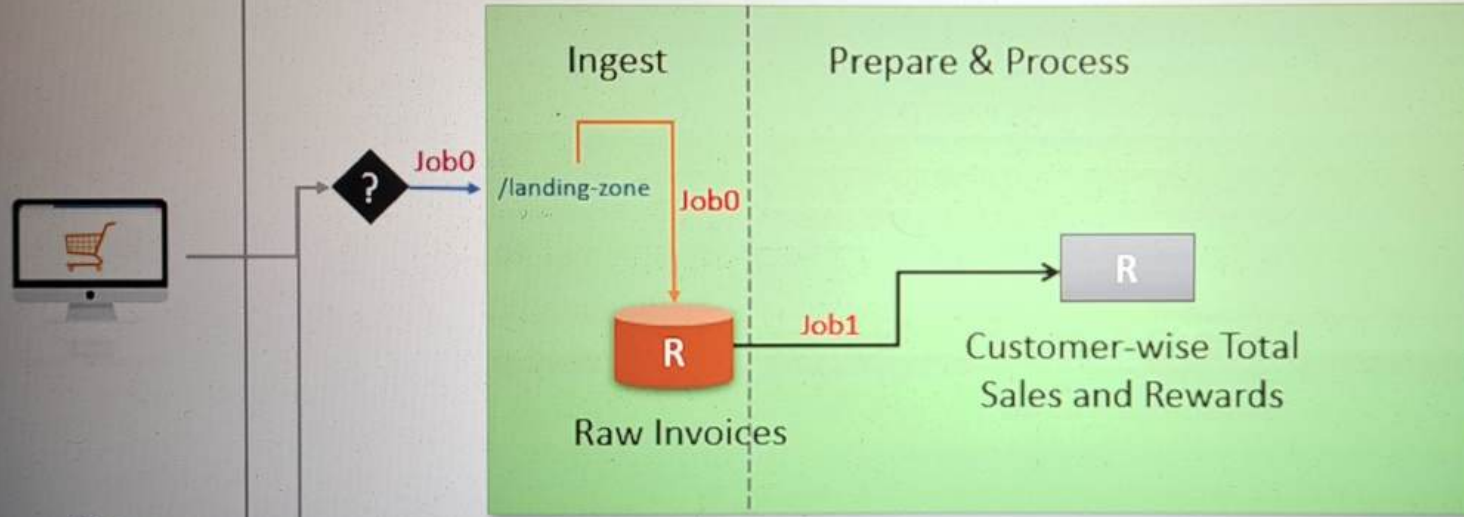


Source System

Data Engineering Platform

Consumer Application



1. How Spark Streaming calculates aggregation
2. What is Spark Streaming State Store
3. Output Modes – Append and Complete



16-streaming-aggregates Python ▾

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Cmd 1

Python

```
class Bronze():~
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def getSchema(self):
        return """InvoiceNumber string, CreatedTime bigint, StoreID string, PosID string, CashierID string,
        CustomerType string, CustomerCardNo string, TotalAmount double, NumberOfItems bigint,
        PaymentMethod string, TaxableAmount double, CGST double, SGST double, CESS double,
        DeliveryType string,
        DeliveryAddress struct<AddressLine string, City string, ContactNumber string, PinCode string,
        State string>,
        InvoiceLineItems array<struct<ItemCode string, ItemDescription string,
        ItemPrice double, ItemQty bigint, TotalValue double>>

    def readInvoices(self):
        from pyspark.sql.functions import input_file_name
        return (spark.readStream
            .format("json")
            .schema(self.getSchema())
            .load(f"{self.base_data_dir}/data/invoices")
            .withColumn("InputFile", input_file_name())
        )

    def process(self):
```



16-streaming-aggregates Python

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```
InvoiceLineItems array<struct<ItemCode string, ItemDescription string,  
    ItemPrice double, ItemQty bigint, TotalValue double>>
```

```
def readInvoices(self):  
    from pyspark.sql.functions import input_file_name  
    return (spark.readStream  
            .format("json")  
            .schema(self.getSchema())  
            .load(f"{self.base_data_dir}/data/invoices")  
            .withColumn("InputFile", input_file_name())  
            )  
  
def process(self):  
    print(f"\nStarting Bronze Stream...", end='')  
    invoicesDF = self.readInvoices()  
    sQuery = ( invoicesDF.writeStream  
              .queryName("bronze-ingestion")  
              .option("checkpointLocation", f"{self.base_data_dir}/checkpoint/invoices_bz")  
              .outputMode("append")  
              .toTable("invoices_bz")  
              )  
    print("Done")  
    return sQuery
```

Shift+Enter to run

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16-streaming-aggregates Python

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```
class Gold():
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def readBronze(self):
        return spark.readStream.table("invoices_bz")

    def getAggregates(self, invoices_df):
        from pyspark.sql.functions import sum, expr
        return (invoices_df.groupBy("CustomerCardNo")
                .agg(sum("TotalAmount").alias("TotalAmount"),
                    sum(expr("TotalAmount*0.02")).alias("TotalPoints"))
                )

    def saveResults(self, results_df):
        print(f"\nStarting Silver Stream...", end='')
        return (results_df.writeStream
                .queryName("gold-update")
                .option("checkpointLocation", f"{self.base_data_dir}/checkpoint/customer_rewards")
                .outputMode("complete")
                .toTable("customer_rewards")
                )
        print("Done")

    def process(self):
        invoices_df = self.readBronze()
```



16-streaming-aggregates Python ▾

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```
def readBronze(self):
    return spark.readStream.table("invoices_bz")

def getAggregates(self, invoices_df):
    from pyspark.sql.functions import sum, expr
    return (invoices_df.groupBy("CustomerCardNo")
            .agg(sum("TotalAmount").alias("TotalAmount"),
                 sum(expr("TotalAmount*0.02")).alias("TotalPoints"))
           )

def saveResults(self, results_df):
    print(f"\nStarting Silver Stream...", end='')
    return (results_df.writeStream
            .queryName("gold-update")
            .option("checkpointLocation", f"{self.base_data_dir}/checkpoint/customer_rewards")
            .outputMode("complete")
            .toTable("customer_rewards")
           )
    print("Done")

def process(self):
    invoices_df = self.readBronze()
    aggregate_df = self.getAggregates(invoices_df)
    sQuery = self.saveResults(aggregate_df)
    return sQuery
```



17-streaming-aggregates-test-suite

Python

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Cmd 1

Python

```
class AggregationTestSuite():
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def cleanTests(self):
        print(f"Starting Cleanup...", end='')
        spark.sql("drop table if exists invoices_bz")
        spark.sql("drop table if exists customer_rewards")
        dbutils.fs.rm("/user/hive/warehouse/invoices_bz", True)
        dbutils.fs.rm("/user/hive/warehouse/customer_rewards", True)

        dbutils.fs.rm(f"{self.base_data_dir}/checkpoint/invoices_bz", True)
        dbutils.fs.rm(f"{self.base_data_dir}/checkpoint/customer_rewards", True)

        dbutils.fs.rm(f"{self.base_data_dir}/data/invoices", True)
        dbutils.fs.mkdirs(f"{self.base_data_dir}/data/invoices")
        print("Done")

    def ingestData(self, itr):
        print(f"\tStarting Ingestion...", end='')
        dbutils.fs.cp(f"{self.base_data_dir}/datasets/invoices/invoices_{itr}.json", f"{self.base_data_dir}/data/invoices_{itr}.json")
        print("Done")

    def assertBronze(self, expected_count):
        print(f"\tStarting Bronze validation...", end='')
```



17-streaming-aggregates-test-suite Python ▾

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```
dbutils.fs.cp(f"{self.base_data_dir}/datasets/invoices/invoices_{itr}.json", f"{self.base_data_dir}/data/invoices/")
print("Done")
```

```
def assertBronze(self, expected_count):
    print(f"\tStarting Bronze validation...", end='')
    actual_count = spark.sql("select count(*) from invoices_bz").collect()[0][0]
    assert expected_count == actual_count, f"Test failed! actual count is {actual_count}"
    print("Done")
```

```
def assertGold(self, expected_value):
    print(f"\tStarting Gold validation...", end='')
    actual_value = spark.sql("select TotalAmount from customer_rewards where CustomerCardNo = '2262471989']").collect()[0][0]
    assert expected_value == actual_value, f"Test failed! actual value is {actual_value}"
    print("Done")
```

```
def waitMicroBatch(self, sleep=30):
    import time
    print(f"\tWaiting for {sleep} seconds...", end='')
    time.sleep(sleep)
    print("Done.")
```

```
def runTests(self):
    self.cleanTests()
    bzStream = Bronze()
    bzQuery = bzStream.process()
    gdStream = Gold()
    gdQuery = gdStream.process()
```



17-streaming-aggregates-test-suite

Python

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Run all

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```
time.sleep(sleep)
```

```
print("Done.")
```

```
def runTests(self):
```

```
    self.cleanTests()
```

```
    bzStream = Bronze()
```

```
    bzQuery = bzStream.process()
```

```
    gdStream = Gp1d()
```

```
    gdQuery = gdStream.process()
```

```
    print("\nTesting first iteration of invoice stream...")
```

```
    self.ingestData(1)
```

```
    self.waitForMicroBatch()
```

```
    self.assertBronze(501)
```

```
    self.assertGold(36859)
```

```
    print("Validation passed.\n")
```

```
    print("\nTesting second iteration of invoice stream...")
```

```
    self.ingestData(2)
```

```
    self.waitForMicroBatch()
```

```
    self.assertBronze(501+500)
```

```
    self.assertGold(36859+20740)
```

```
    print("Validation passed.\n")
```

```
    print("\nTesting second iteration of invoice stream...")
```

```
    self.ingestData(3)
```

```
    self.waitForMicroBatch()
```

```
    self.assertBronze(501+500+590)
```



17-streaming-aggregates-test-suite

Python ▾

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```
self.assertGold(36859)
print("Validation passed.\n")

print("\nTesting second iteration of invoice stream...")
self.ingestData(2)
self.waitForMicroBatch()
self.assertBronze(501+500)
self.assertGold(36859+20740)
print("Validation passed.\n")

print("\nTesting second iteration of invoice stream...")
self.ingestData(3)
self.waitForMicroBatch()
self.assertBronze(501+500+590)
self.assertGold(36859+20740+31959)
print("Validation passed.\n")

bzQuery.stop()
gdQuery.stop()
```

Shift+Enter to run

Shift+Ctrl+Enter to run selected text



17-streaming-aggregates-test-suite Python ▾

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Interrupt

my cluster ▾

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```
guquery.stop()
```

Command took 0.13 seconds -- by prashantkumarpandey@gmail.com at 11/1/2023, 1:46:42 PM on my cluster

Cmd 2

```
aTS = AggregationTestSuite()  
aTS.runTests()
```

Running command...

Starting Cleanup...

Shift+Enter to run

Shift+Ctrl+Enter to run selected text



17-streaming-aggregates-test-suite Python ▾

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▶ Run all

● my cluster ▾

Share

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▶ (26) Spark Jobs



Starting Cleanup...Done

Starting Bronze Stream...Done

Starting Silver Stream...

Testing first iteration of invoice stream...

Starting Ingestion...Done

Waiting for 30 seconds...Done.

Starting Bronze validation...Done

Starting Gold validation...Done

Validation passed.

Testing second iteration of invoice stream...

Starting Ingestion...Done

Waiting for 30 seconds...Done.

Starting Bronze validation...Done

Starting Gold validation...Done

Validation passed.

Command took 2.20 minutes -- by prashantkumarpandey@gmail.com at 11/1/2023, 1:47:47 PM on my cluster

Shift+Enter to run

Shift+Ctrl+Enter to run selected text



CustomerCardNo	InvoiceNumber	CreatedTime	TotalAmount	InputFile	Micro batch Totals	State Store
2262471989	97653721	1595689063675	3850	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	70630270	1595689081744	7659	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	12361996	1595689187238	11285	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	98939606	1595689189886	5737	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	30231439	1595689193881	570	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	96055254	1595689194520	2428	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	10025390	1595689195570	2742	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	47495165	1595689197375	500	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.json		
2262471989	57160284	1595689201304	2088	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_1.js	36859	36859
2262471989	60255959	1595689264075	2442	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_2.json		
2262471989	86910195	1595689267533	6991	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_2.json		
2262471989	67609664	1595689269995	7302	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_2.json		
2262471989	31431291	1595689270396	4005	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_2.js	20740	57599
2262471989	98611923	1595689272419	6386	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.json		
2262471989	3294604	1595689279888	1417	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.json		
2262471989	28927842	1595689298279	7220	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.json		
2262471989	43729844	1595689302400	4870	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.json		
2262471989	35313766	1595689304188	7040	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.json		
2262471989	99674780	1595689307129	3086	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.json		
2262471989	43973698	1595689307899	1940	dbfs:/FileStore/data_spark_streaming_scholarnest/data/invoices/invoices_3.js	31959	89558

CustomerCardNo	TotalAmount	TotalPoints
1 2262471989	89558	

Problem - Complete Output Mode

